

Muhammad Rehman Ashraf

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SUMMARY

- AI Software Engineer with **2+ years of experience** in Computer Vision and NLP, specializing in building and deploying end-to-end AI systems.
- Experienced in designing models from scratch, training pipelines, and deploying scalable AI applications using modern backend frameworks and cloud platforms.
- Experienced in deploying end-to-end AI systems with cloud infrastructure and **CI/CD pipelines** for scalable and reliable delivery.

EDUCATION

Pakistan Institute of Engineering and Applied Sciences (PIEAS)

Islamabad, Pakistan

Bachelor of Computer Science

Sep 2022 – Sep 2026

- Relevant Coursework: Data Structures, Algorithms, Machine Learning, AI, NLP, Computer Vision, Linear Algebra, OOP, Database Systems, IoT, Probability, Statistics, and Optimization

RESEARCH & INTERNSHIP EXPERIENCE

Undergraduate Research Assistant

Dec 2022 – Present

PIEAS University

Islamabad, Pakistan

- Completed an internship in web development from High Tech Software House & Training Center [Certificate]
- AI/Deep Learning/Computer Vision badge [Certificate]
- Participated in **AI Hackathon at DevCon 26'** — demonstrated innovation, technical expertise, and teamwork in building intelligent solutions [Certificate]
- Developed AI pipelines for **3D human motion capture** and posture analysis using SMPL, SMPLX, OpenCV and MediaPipe.
- Developed AI pipelines for **object detection and tracking, traffic monitoring system, and mall customer monitoring system**.
- Built full-stack AI systems using React.js, Node.js, FastAPI, and SQL/NoSQL databases.
- Handled real-world noisy datasets, preprocessing, validation, and deployment workflows.

SELECTED PROJECTS

JobHunting-Agent — AI Career Assistant | *Python, FastAPI, Agentic AI, Gemini, Tavily* | GitHub

2026

- Built an agentic AI system that automates job search, CV analysis, and tailored application generation.
- Integrated LLM reasoning with real-time search for intelligent job matching.

KongoSnap — AI Learning App | *React.js, Node.js, React Native* | GitHub

2026

- Developed AI-powered app generating quizzes and explanations from images.

DocMind — RAG Chat App | *React.js, FastAPI, PostgreSQL* | GitHub

2026

- Built document chat system with embeddings and AI responses.

Image Captioning Model | *PyTorch, AWS EC2* | GitHub

2026

- Developed CNN+Transformer model for image-to-text generation.

Traffic Analysis System | *YOLO, OpenCV, FastAPI* | GitHub

2025

- Built real-time traffic monitoring system with detection, tracking, and analytics.

YOLO ByteTrack Video Heatmap API | *YOLOv11, FastAPI* | GitHub

2025

- Implemented multi-object tracking and heatmap generation for surveillance insights.

ESP8266 IoT Sketches | *Arduino, ESP8266, MQTT, GPS, MPU6050, Node-RED* | GitHub

2025

- Developed four embedded IoT sketches: GPS live tracking, MQTT-based LED control over HiveMQ broker, MPU6050 fall detection with HTTP alerts, and periodic sensor data posting to a Node-RED dashboard.
- Integrated SoftwareSerial, I2C, WiFi, and pub/sub protocols on resource-constrained hardware.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C/C++, C#

Frontend: React.js, React Native, Angular.js, HTML, CSS, JavaScript, Tailwind CSS

Backend: FastAPI, Node.js, NestJS, ASP.NET, REST APIs

Mobile: React Native, Basic Android Development, Advanced Backend

Databases: MySQL, PostgreSQL, MongoDB (NoSQL)

AI/ML: Computer Vision, NLP, Deep Learning, Transformers, Object Detection, Pose Detection, Segmentation, Vision-Language Models, Fine-tune LLM with LoRA, n8n, LangChain, CrewAI

Libraries: PyTorch, TensorFlow, OpenCV, MediaPipe, NumPy, Pandas, Ultralytics

IoT & Embedded: ESP8266, Arduino, MQTT (HiveMQ/PubSubClient), GPS (TinyGPSPlus), MPU6050, I2C, SoftwareSerial, Node-RED, HTTP/REST on microcontrollers

Cloud & DevOps: AWS, Azure, Docker, CI/CD Pipelines, Vercel, Render

Tools & Collaboration: Git, GitHub, VS Code, Visual Studio, Cursor, Jupyter Notebook, Agile Workflows, Team Collaboration, Technical Documentation